Computer Science - Graduate Level Assessment 20 Total Marks: 40 Answer all questions.

Multiple Choice (10 marks)

1. Which of the following statements about a remote procedure call is true?
   (a) It is used to call procedures with addresses that are farther than 2^16 bytes away.
   (b) It cannot return a value.
   (c) It cannot pass parameters by reference.
   (d) It cannot call procedures implemented in a different language.
2. A sender must not be able to deny sending a message that was sent, is known as
   (a) Message Nonrepudiation
   (b) Message Integrity
   (c) Message Condentiality
   (d) Message Sending
3. ______________________ is the central node of 802.11 wireless operations.
   (a) WPA
   (b) Access Point
   (c) WAP
   (d) Access Port
4. Which of the following comes closest to being a perfectly secure encryption scheme?
   (a) The Caesar Cipher, a substitution cipher
   (b) DES (Data Encryption Standard), a symmetric-key algorithm
   (c) Enigma, a transposition cipher
   (d) One-time pad
5. The language {ww | w in (0 + 1)*} is
   (a) not accepted by any Turing machine
   (b) accepted by some Turing machine, but by no pushdown automaton
   (c) accepted by some pushdown automaton, but not context-free
   (d) context-free, but not regular

True / False (10 marks)

Answer True or False

6. True or False: Authorization aims to restrict what operations and data a user can access within a system.
7. True or False: The four Primary Security Principles related to messages are Confidentiality, Integrity, Non-repudiation, and Authentication.
8. True or False: A queue is a sequential segment of memory used for temporarily holding data before processing, similar to a buffer.
9. True or False: Message confidentiality guarantees the integrity of the data transmitted between the sender and receiver.
10. True or False: Merge sort has both average-case and worst-case time complexities of O(n log n).

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. Write a function to find the minimum product from the pairs of tuples within a given list.
12. Write a python function to count the number of lists in a given number of lists.

End of Paper